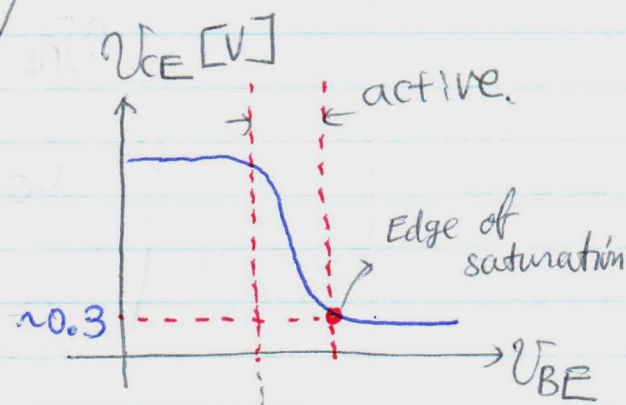
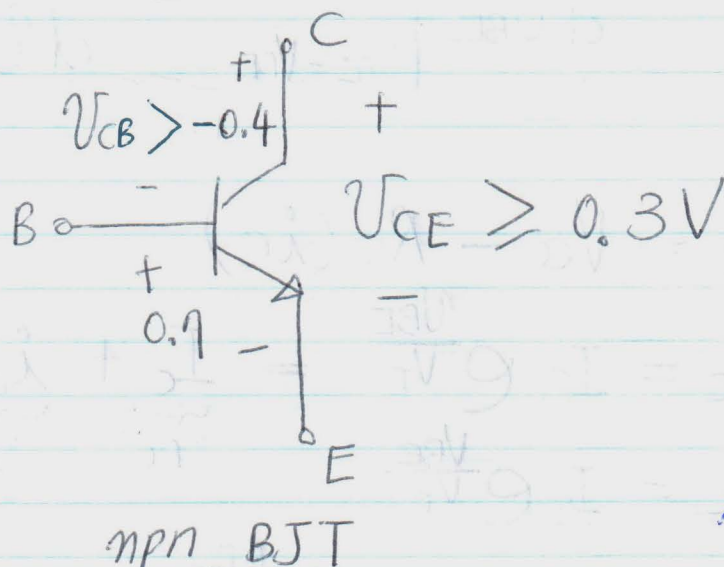


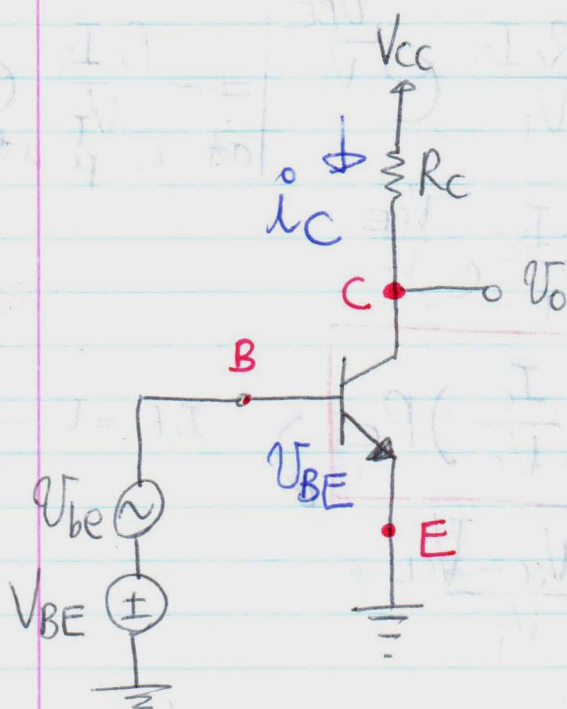
< BJT Amplifier >

1. How to check if a BJT is in an active mode?



If $V_{be} = 0.4V$, then $V_{ce} \geq 0V$

2. Small-signal voltage gain.



$$V_{BE} = \underbrace{V_{BE}}_{\text{"DC"}} + \underbrace{v_{be}}_{\text{"AC"}}$$

= input signal

$$V_o = V_{CC} - R_C(i_C)$$

= output signal

small-signal voltage gain, A_v ,

$$A_v \equiv \left. \frac{dV_o}{dV_{BE}} \right|_{V_{BE}=V_{BE}} = \left. \frac{dV_{CE}}{dV_{BE}} \right|_{\text{at Q-pt. int.}}$$

where,

$$\begin{cases} V_{CE} = V_{CC} - R_C(i_C) \\ i_C = I_S e^{\frac{V_{BE}}{V_T}} = \underbrace{I_C}_{\text{"DC"}} + \underbrace{i_c}_{\text{"AC"}} \\ I_C = I_S e^{\frac{V_{BE}}{V_T}} \end{cases}$$

$$V_{CE} = V_{CC} - R_C I_S e^{\frac{V_{BE}}{V_T}}$$

$$\left. \frac{dV_{CE}}{dV_{BE}} \right|_{\text{at Q-point}} = - \frac{R_C I_S}{V_T} e^{\frac{V_{BE}}{V_T}}$$

$$= - \frac{R_C I_S}{V_T} e^{\frac{V_{BE}}{V_T}}$$

$$= - \left(\frac{I_C}{V_T} \right) R_C$$

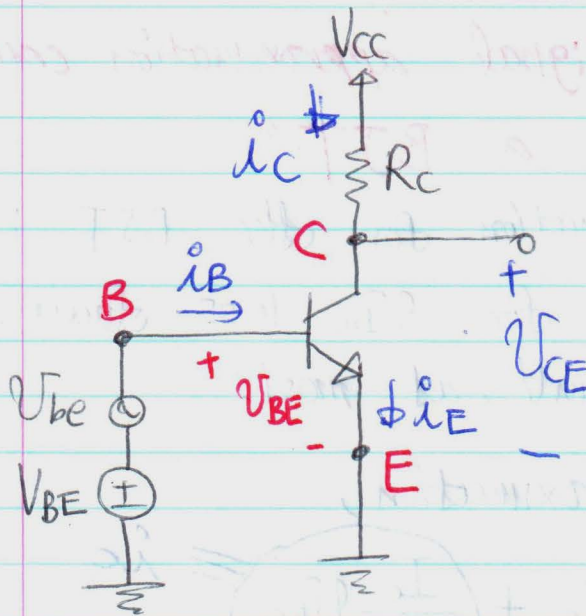
$$I_C R_C = V_{CC} - V_{CE}$$

$$= - \frac{V_{CC} - V_{CE}}{V_T}$$

V_T = thermal voltage $\rightarrow \sim 25 \text{ mV}$ @ room temp.

4. Small-signal operation and models.

Let's consider the small-signal operation of the BJT and develop small-signal equivalent circuit models that represent its operation at a given **bias point**.



① DC Bias point

$$I_C = I_S e^{V_{BE}/V_T}$$

$$I_E = I_C / \alpha$$

$$I_B = I_C / \beta$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$\alpha = \frac{\beta}{1 + \beta}$$

② Collector current & the transconductance

$$i_C = I_S e^{\frac{v_{BE}}{V_T}}$$

$$= I_S e^{\frac{V_{BE} + v_{be}}{V_T}} = I_S e^{\frac{V_{BE}}{V_T}} e^{\frac{v_{be}}{V_T}}$$

$$= I_C e^{\frac{v_{be}}{V_T}}$$

$$\approx I_C \left(1 + \frac{v_{be}}{V_T}\right) \quad \left. \vphantom{\frac{v_{be}}{V_T}} \right\} \text{ if } v_{be} \ll V_T,$$

Here we have expanded the exponential in a series and retained only the first two terms. That is, we have assumed that

$$V_{be} \ll V_T$$

"small-signal approximation condition for a BJT"

$\Rightarrow$ this approximation for the BJT is valid only for V_{be} less than 5 mV $\sim$ 10 mV, at most.

Under this approximation,

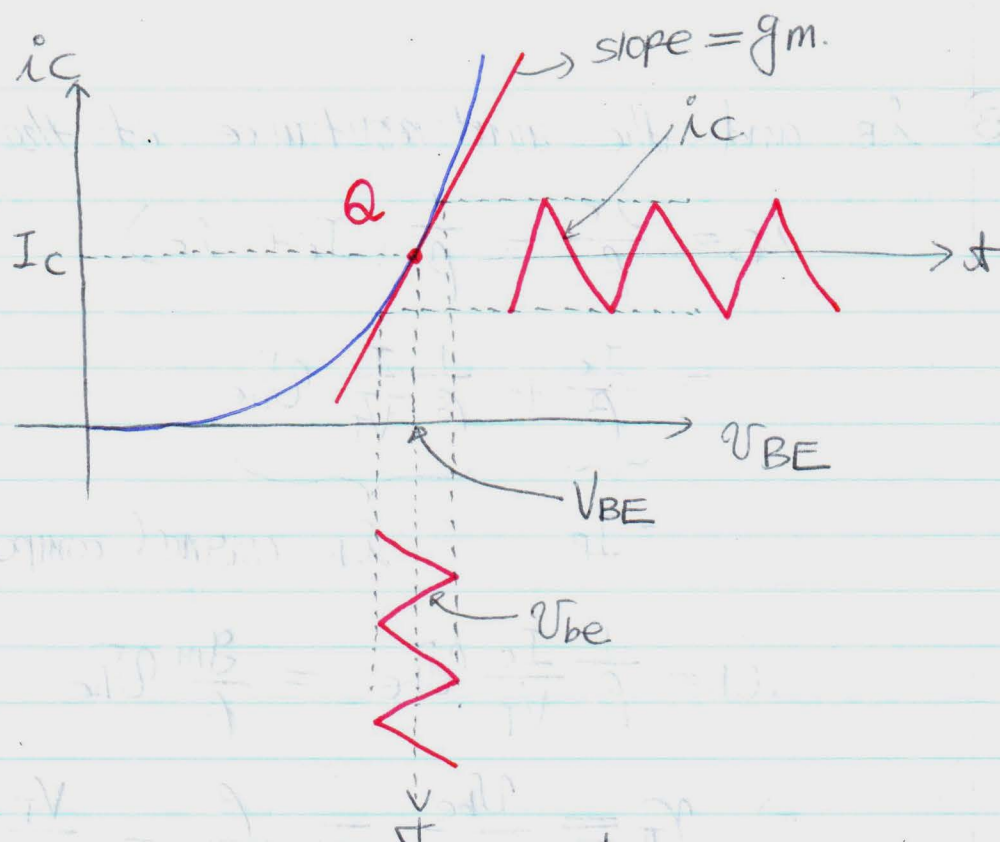
$$\dot{i}_c = I_c + \left(\frac{I_c}{V_T} V_{be} \right) \equiv \dot{i}_c$$

$\Downarrow$
dc bias component

$\Downarrow$
small-signal component

$$\dot{i}_c = \frac{I_c}{V_T} V_{be} = g_m V_{be}$$

$$g_m \text{ (transconductance)} \equiv \frac{\dot{i}_c}{V_{be}} = \frac{I_c}{V_T}$$



$$g_m = \left. \frac{\partial i_c}{\partial V_{BE}} \right|_{i_c = I_C \text{ (}\therefore \text{Q-point)}}.$$

The small-signal approximation implies keeping the signal amplitude sufficiently small that operation is restricted to an **almost-linear segment** of the $i_c - V_{BE}$ exponential curve.

$\Rightarrow$ Increasing the signal amplitude will result in the collector current having components nonlinearly related to v_{be} .

③ i_B and the input resistance at the Base.

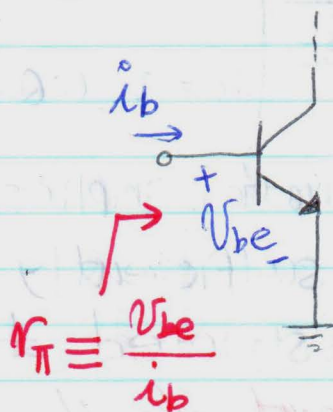
$$i_B = \frac{i_C}{\beta} = \frac{1}{\beta} (I_C + \underline{i_c})$$

$$= \frac{I_C}{\beta} + \frac{1}{\beta} \frac{I_C}{V_T} v_{be}$$

$$= I_B \quad \underbrace{\quad}_{\underline{i_b} \text{ (signal component)}}$$

$$\underline{i_b} = \frac{1}{\beta} \frac{I_C}{V_T} \underline{v_{be}} = \frac{g_m}{\beta} \underline{v_{be}}$$

$$\Rightarrow r_{\pi} \equiv \frac{v_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$



④ i_E and the input resistance at the Emitter

$$i_E = \frac{i_C}{\alpha} = \frac{I_C}{\alpha} + \frac{\underline{i_c}}{\alpha} = I_E + \underline{i_e}$$

$$\underline{i_e} = \frac{\underline{i_c}}{\alpha} = \frac{I_c}{\alpha V_T} v_{be} = \frac{I_E}{V_T} \underline{v_{be}}$$

$$\boxed{r_e \text{ (emitter resistance)}} \equiv \frac{\underline{v_{be}}}{\underline{i_e}} = \frac{V_T}{I_E}$$

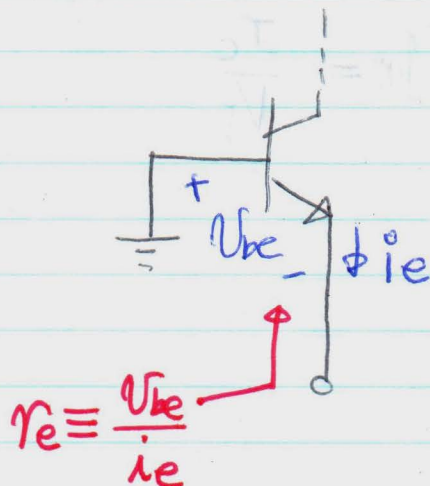
$$= \frac{\alpha V_T}{I_c} = \boxed{\frac{\alpha}{g_m}} \simeq \frac{1}{g_m}$$

$$v_{be} = i_b r_{\pi} = i_e r_e$$

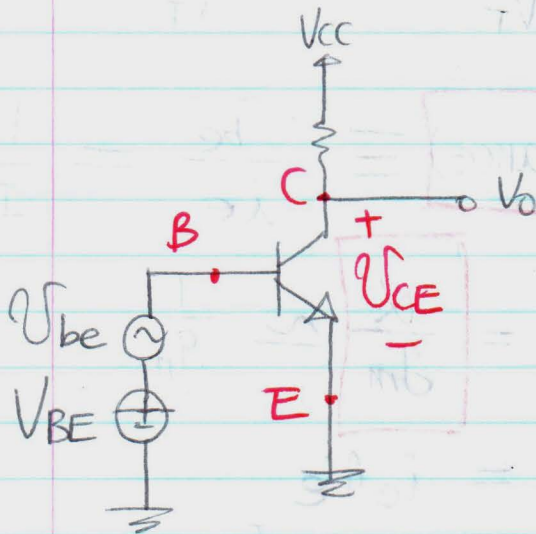
$$\text{thus, } r_{\pi} = \frac{i_e}{i_b} r_e = \frac{\frac{I_c}{\alpha V_T} v_{be}}{\frac{g_m}{\beta} v_{be}} r_e$$

$$r_{\pi} = \frac{\beta}{\alpha} r_e = \frac{\beta}{\frac{\beta}{1+\beta}} r_e = (1+\beta) r_e$$

$$\Rightarrow \boxed{r_{\pi} = (1+\beta) r_e}$$



⑤ Voltage Gain



$$\begin{aligned}
 V_{CE} &= V_{CC} - i_c R_C \\
 &= V_{CC} - (I_C + i_c) R_C \\
 &= (V_{CC} - I_C R_C) - i_c R_C \\
 &= V_{CE} - i_c R_C \\
 &= V_{CE} + \underline{V_{ce}}
 \end{aligned}$$

thus,

$$\begin{aligned}
 \underline{V_{ce}} &= -i_c R_C = -g_m \underline{V_{be}} R_C \\
 &= -(g_m R_C) \underline{V_{be}}
 \end{aligned}$$

$$A_v \equiv \frac{V_{ce}}{V_{be}} = -g_m R_C$$

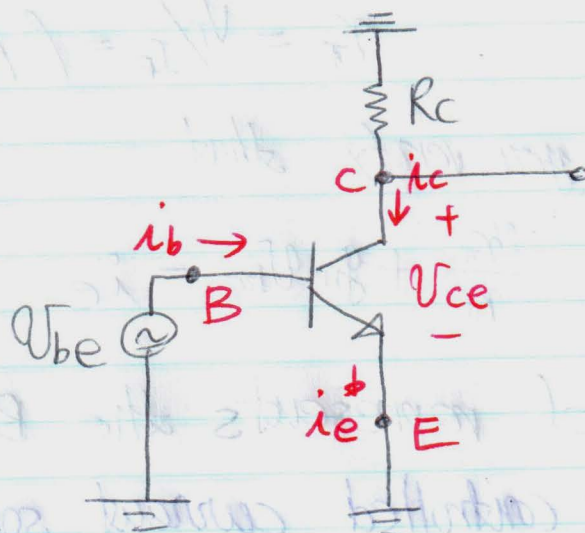
$$g_m = \frac{I_C}{V_T}$$

⑥ Separating the Signal & the DC components.

every current and voltage in the amplifier circuit is composed of two components:

dc component and ac component
signal

⇒ representation of the signal operation of the BJT can be obtained by eliminating the **dc source**.



$$i_c = g_m V_{be}$$

$$i_b = V_{be} / r_{\pi}$$

$$i_e = \frac{V_{be}}{r_e}$$

$$r_e = \frac{\alpha}{g_m}$$

$$g_m = \frac{I_c}{V_T}$$

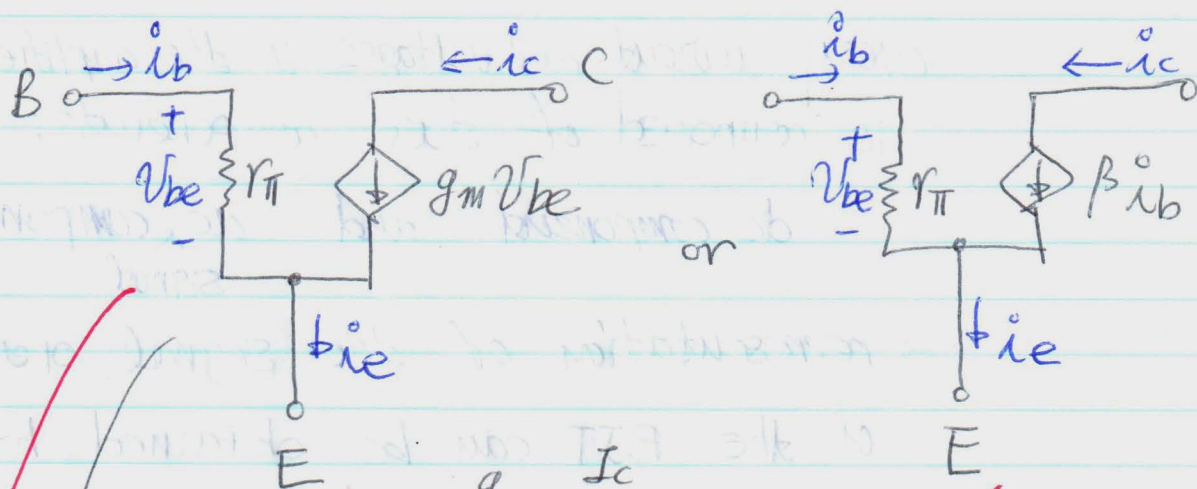
$$r_{\pi} = \frac{\beta}{g_m}$$

$$r_{\pi} = (1 + \beta) r_e$$

$$g_m r_{\pi} = \beta$$

dc sources (V_{BE} & V_{CC}) eliminated (short-circuited).

① The hybrid- π equivalent-circuit model.



$$g_m = \frac{I_c}{V_T}$$

$$r_{\pi} = V_T / I_B = \beta / g_m$$

Can you verify that

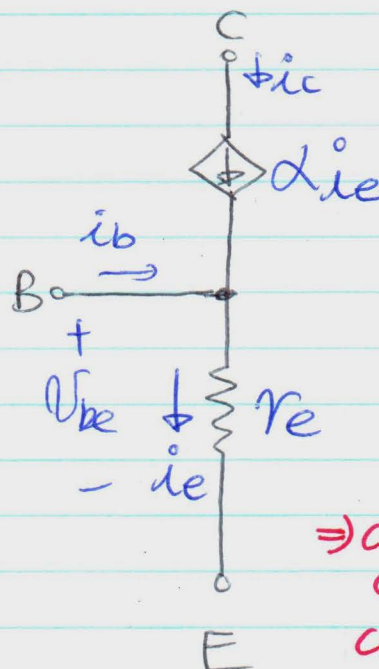
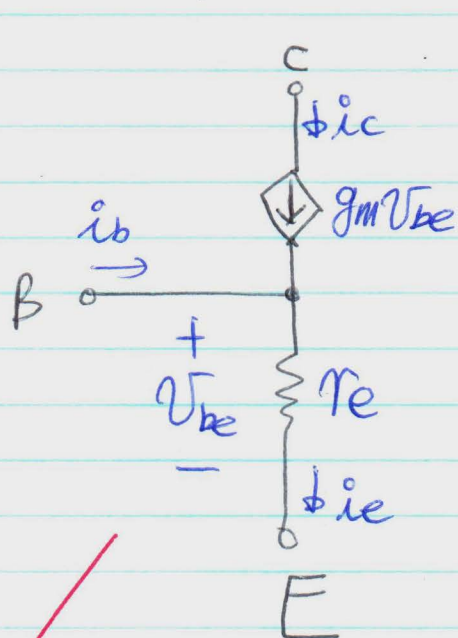
$$\frac{V_{be}}{r_{\pi}} + g_m V_{be} = i_e \quad ?$$

This model represents the BJT as a voltage controlled current source.

current controlled current source

⑧ The T-model.

Although the hybrid- π model can be used to carry out small-signal analysis of any transistor circuit, there are situations in which an alternative model, known as a T-model, is much more convenient.



$\Rightarrow$ current controlled current source.

Can you verify that

$$i_b + g_m V_{be} = i_e = \frac{V_{be}}{r_e} \quad ?$$

$\Rightarrow$ This model represents the BJT as a voltage controlled current source

⑨ Small-signal model parameters

$$i_c = \frac{I_c}{V_T} v_{be} \Rightarrow g_m \equiv \frac{i_c}{v_{be}} = \frac{I_c}{V_T}$$

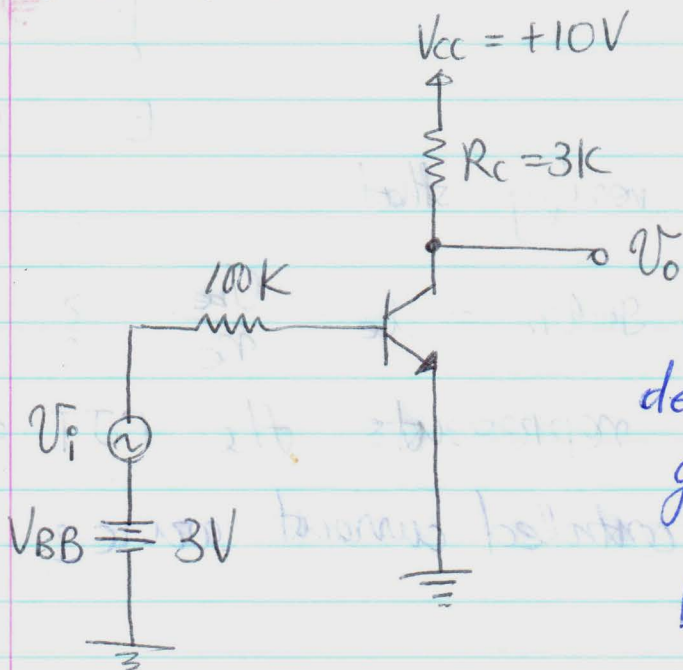
$$i_b = \frac{g_m}{\beta} v_{be} \Rightarrow r_{\pi} \equiv \frac{v_{be}}{i_b} = \frac{\beta}{g_m}$$

$$i_e = \frac{I_E}{V_T} v_{be} \Rightarrow r_e \equiv \frac{v_{be}}{i_e} = \frac{\alpha}{g_m} \approx \frac{1}{g_m}$$

$$\text{also, } r_{\pi} = (1 + \beta) r_e$$

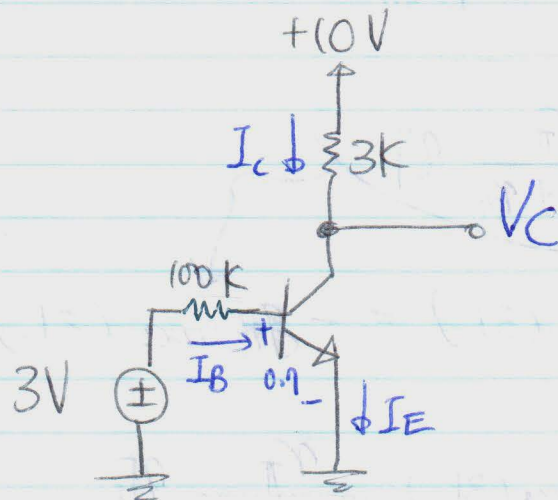
$$A_v \equiv \frac{v_o}{v_{be}} = -g_m R_C = -\frac{I_c}{V_T} R_C$$

⑩ Example.



determine the voltage gain, v_o/v_i
 $\beta = 100$.

i) DC-bias analysis



$$I_B = \frac{3 - 0.7}{100k} = 0.023 \text{ mA}$$

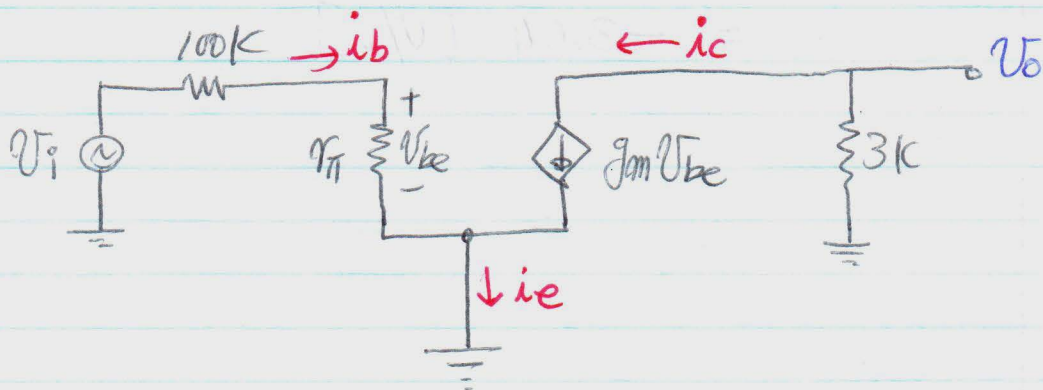
$$I_C = \beta \cdot I_B = 2.3 \text{ mA}$$

$$I_E = I_B + I_C = 2.323 \text{ mA}$$

$$V_C = 10 - (3k) I_C = + 3.1 \text{ V}$$

$\Rightarrow V_C > 0.3 \text{ V} \Rightarrow \text{"active mode"}$

ii) small-signal analysis



$$g_m = \frac{I_C}{V_T} = \frac{2.3 \text{ mA}}{0.025} = 92 \text{ mA/V}$$

$$r_{\pi} = \frac{\beta}{g_m} = \frac{100}{92 \text{ m}} = 1.09 \text{ k}\Omega$$

$$v_{be} = \frac{r_{\pi}}{100 \text{ k} + r_{\pi}} v_i$$

$$v_o = -i_c (3 \text{ k}) = -g_m v_{be} (3 \text{ k})$$

$$\Rightarrow v_o = -g_m (3 \text{ k}) \frac{r_{\pi}}{100 \text{ k} + r_{\pi}} v_i$$

$$\Rightarrow A_v = \frac{v_o}{v_i} = -g_m (3 \text{ k}) \frac{r_{\pi}}{100 \text{ k} + r_{\pi}}$$

$$= -(92 \text{ m})(3 \text{ k}) \frac{1.09 \text{ k}}{100 \text{ k} + 1.09 \text{ k}}$$

$$= -3.04 \text{ [V/V]}$$